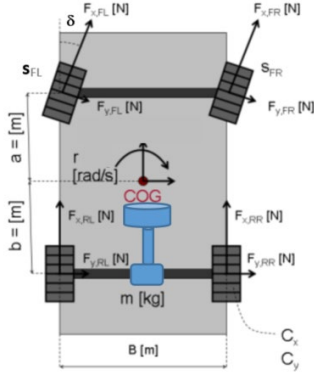


RaceCar Matematik Modeli: Race

Car arka akstan itişli bir araç modeline sahiptir. Modelde steering sistemindeki sürtünme etkisi ihmal edilmiştir. Ayrıca iç ortamda ve düşük hızlarda hareket söz konusu olduğu için hava direnci ihmal edilmiştir.



$$\frac{d}{dt} \begin{bmatrix} X \\ Y \\ \psi \\ v_x \\ v_y \\ r \\ \delta \\ \omega_\delta \\ \omega_W \end{bmatrix} = \begin{bmatrix} v_x \cos \psi - v_y \sin \psi \\ v_x \sin \psi + v_y \cos \psi \\ r \\ v_y r + \frac{2F_{x_f}}{m} - \frac{2F_{y_f} \sin \delta}{m} \\ -v_x r + \frac{2F_{y_f} \cos \delta}{m} + \frac{2F_{y_r}}{m} \\ \frac{a 2F_{y_f} \cos \delta}{I_z} - \frac{b 2F_{y_r}}{I_z} \\ \omega_\delta \\ -\frac{B_{st}}{J_{st}} \omega_\delta - \frac{K_{st}}{J_{st}} \delta \\ -\frac{B_L}{J_W} \omega_W - \frac{2F_{x_f} r_W}{J_W} \end{bmatrix} + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{J_{st}} & 0 \\ 0 & \frac{1}{J_W} \end{bmatrix} \begin{bmatrix} T_{st} \\ T_W \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\frac{1}{m} \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} \begin{bmatrix} F_{x_{res}} \\ d_c \end{bmatrix}$$

$$\dot{x}_{RC}(t) = f_{RC}(x_{RC}(t)) + B_C u_C(t) + k_C d_C(t) \quad (1)$$

$F_{resx}(t) = mg \sin \theta + mg f_{rx} \cos \theta + k_{aex} v_x^2$ ($mg \sin \theta$: gravity, $mg f_{rx} \cos \theta$: longitudinal rolling and $k_{aex} v_x^2$ aerodynamic reaction force components). Where, X and Y inertial frame positions ψ : heading angle; v_x and v_y : longitudinal and lateral velocities of CG; r : heading rate; δ : steering angle; ω_δ : steering rate; ω_W : wheel angular speed; m : vehicle total mass; $I_z = \left(\frac{a+b}{2}\right)^2 m$; I_z : total traction inertia with respect to the z axis; J_{st} : total steering inertia, J_W : total wheel inertia B_{st}, B_W : total steering and traction system viscous coefficients, respectively; thrust (drive) force.